LAB 02

Introduction to **Object-Oriented Programming**

{OOP} **Basics in Java**

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Lab02 Exercises

❖ Lab02.1. Hello One's Name

Implement an application Java that displays "Hello <input_name>!" where <input_name> represents user input from keyboard. Example:

Console:

```
Input your name: Visal
Hello Visal!
```

❖ Lab02.2. Currency converter

Write a program in Java to money in Riels to Dollar. Suppose that conversion rate is 1$ = 4000₹. Example:

Console:

```
Program for converting money in Riels to Dollars.
Please input conversion rate of 1 dollar in Riels: 4000
Please input money to exchange in Riels: 2200

2200 RIELS = 0.55 USD
```

Lab02 Exercises

❖ Lab02.3. Health Screening Questions

Write a program to ask questions for checking Health regarding to Covid-19

Console:

```
==== Covid-19 Screen Health ===
Q1. Feeling feverish and/or having chills? Yes
Q2. Has there been any use of fever reducing medication within the last 24 hours not due to another health
condition? No
Q3. A new cough that is not due to another health condition? No
Q4. New chills that are not due to another health condition? No
Q5. A new sore throat that is not due to another health condition? No
Q6. A new loss of taste or smell? Yes
Q7. Have you had a positive test for the virus that causes COVID-19
disease within the past 10 days? No
Q8. In the past 14 days, have you had close contact (within about 6 feet for 15 minutes or more) with someone with
suspected or confirmed COVID-19? Yes
================================================
| Questions             | Answers               |
================================================
| Q1                    | Yes                   |
------------------------------------------------
| Q2                    | No                    |
------------------------------------------------
| Q3                    | No                    |
------------------------------------------------
| Q4                    | No                    |
------------------------------------------------
| Q5                    | No                    |
------------------------------------------------
| Q6                    | Yes                   |
------------------------------------------------
| Q7                    | No                    |
------------------------------------------------
| Q8                    | Yes                   |
```

Lab02 Exercises

❖ Lab02.4. Seconds to Time

Implement an application Java that take a given number of seconds for converting into time with format `Hours:Minutes:Seconds`. Example:

Console:

```
Input number of seconds: 1259
Time corresponding to 1259 seconds is:: 00:20:59
```

❖ Lab02.5. Equation

Write a program in Java to calculate value of **x** in a formula below:

$$\frac{1}{x} = \frac{1}{y} + \frac{1}{z}$$

Where **y** and **z** are given by user from keyboard. Suppose that **x**, **y**, **z** are non-zero.
Example:

Console:

```
Program for calculating equation 1/x = 1/y + 1/z to find value of x
Please input y: 2.25
Please input z: 2
Result x = 1.0588236
```

Lab02 Exercises

❖ Lab02.6. Prime Number

Prime number is positive number greater than 2 and divisible only to its own and 1.
Implement an application Java that let user input a number then determine if it is a prime number

Example 01. Console:

```
Input a number to check whether it is a prime number: 29
29 is a prime number.
```

Example 02. Console:

```
Input a number to check whether it is a prime number: 57
57 is not a prime number, because it is divisible by 3.
```

❖ Lab02.7. Character counter

Write a program using to count number of Vowel, Consonant and Space:

Console:

```
Input a sentence: Hello classmate
Count of vowel is: 5
Count of consonant is: 9
Count of space is: 1
```

Lab02 Exercises

❖ Lab02.8. Minimum value

Write a program to find a minimum value among A, B, C, D and E

Console:

```
Input value of A: 34
Input value of B: 77
Input value of C: 12
Input value of D: 99
Input value of E: 12

The smallest number among A, B, C, D, and E is: 12
```

❖ Lab02.9. Maximum value

Write a program to find a maximum value among all values in an Array. Users can input values in Array as many as they want.

Console:

```
Input value in Array at index 0: 1
Do you want to add more (y/n)?: y
Input value in Array at index 1: 7
Do you want to add more (y/n)?: y
Input value in Array at index 2: 99
Do you want to add more (y/n)?: y
Input value in Array at index 3: 76
Do you want to add more (y/n)?: n

The biggest number among in Array is: 99
```

Lab02 Exercises

❖ Lab02.10. Math Menu

Write a program to choose menu and perform calculation based on options provided; if user choose an incorrection option, program still keep asking to re-input a correct option number:

Example #1:

Console:

```
=== Math Menu ===
1. Addition (+)
2. Substruction (-)
3. Multiplication (x)
4. Division (/)
5. Quit

Please choose an option: 1
```

Console:

```
=== Addition (+)===
Please input value A: 3
Please input value B: 5

Answer of A + B = 8

Do you want to continue? If no go to Menu.
Input your answer (y/n): Y
```

Example #2:

Console:

```
=== Math Menu ===
1. Addition (+)
2. Substruction (-)
3. Multiplication (x)
4. Division (/)
5. Quit

Please choose an option: 6
```

Console:

```
Please choose an option again.
=== Math Menu ===
1. Addition (+)
2. Substruction (-)
3. Multiplication (x)
4. Division (/)
5. Quit

Please choose an option:
```

Lab02 Exercises

❖ Lab02.11. Class and Object

Create a model following objects:

- ID Card
- Facebook Post
- Phone
- Book

1. Implement a **Java Class** with **at least 5 attributes** defined
2. Create **at least 3 objects** with values assigned
3. Create **an array** and put all objects to that array

Follow the example in the figure to do your exercise:

```java
class Student {
    int sid;
    String name;
    char gender;
    int age;
    double score;
}

public class Test{
     public static void main(String []args){
        Student stu1 = new Student();
        Student stu2 = new Student();
        Student stu3 = new Student();
        stu1.sid = 111;
        stu1.name = "Tola";
        stu1.gender = 'M';
        stu1.age = 17;
        stu1.score = 80;

        stu2.sid = 222;
        stu2.name = "Kompheak";
        stu2.gender = 'M';
        stu2.age = 16;
        stu2.score = 60;

        stu3.sid = 333;
        stu3.name = "Makara";
        stu3.gender = 'F';
        stu3.age = 19;
        stu3.score = 90;

        Student[] stuArr = new Student[3];
        stuArr[0] = stu1;
        stuArr[1] = stu2;
        stuArr[2] = stu3;
     }
}
```

Good luck 👌

✋ How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don’t zip the files for submission